

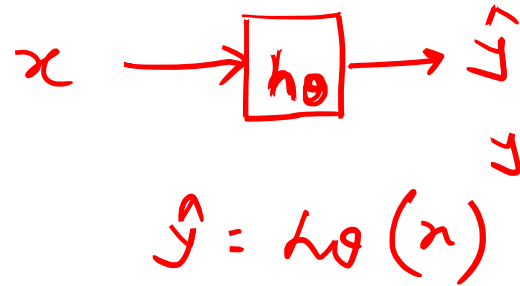
Learning Paradigms

Notations

- a, b, c Scalar (integer or real)
- $\mathbf{x}, \mathbf{y}, \mathbf{z}$ Vector (bold-font, lower case)
- $\mathbf{A}, \mathbf{B}, \mathbf{C}$ Matrix (bold-font, upper-case)
- $\mathbf{A}, \mathbf{B}, \mathbf{C}$ Tensor ((bold-font, upper-case)
- X, Y, Z Random variable (normal font, upper-case)
- $a \in \mathcal{A}$ Set membership: a is member of set $\mathcal{A}$
- $|\mathcal{A}|$ Cardinality: number of items in set $\mathcal{A}$
- $\|\mathbf{v}\|$ Norm of vector $\mathbf{v}$
- $\mathbf{u} \cdot \mathbf{v}$ or $\langle \mathbf{u}, \mathbf{v} \rangle$ Dot product of vectors $\mathbf{u}$ and $\mathbf{v}$
- $\mathbb{R}$ Set of real numbers
- $\mathbb{R}^n$ Real numbers space of dimension n
- $y = f(x)$ or $x \mapsto f(x)$ Function (map): assign a unique value $f(x)$ to each input value x
- $f: \mathbb{R}^n \rightarrow \mathbb{R}$ Function (map): map an n -dimensional vector into a scalar

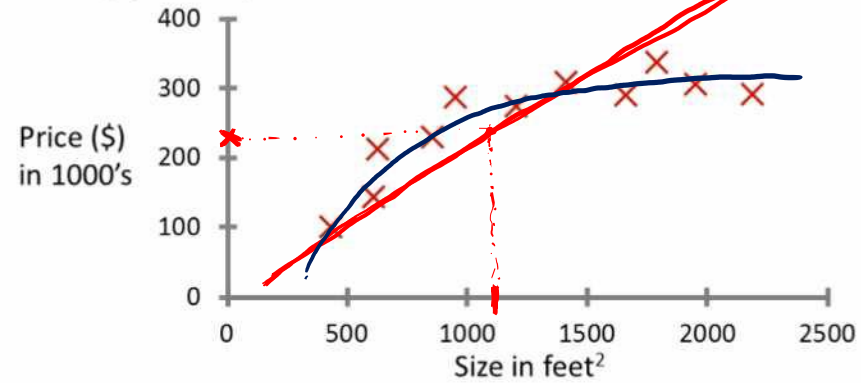
Supervised Learning

- Given: Labelled training data $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$
- Objective: Learn a mapping function $(h : x \rightarrow y)$ to predict output y from new input x .
- The output y can be either
 - i. real valued
 - ii. discrete valued

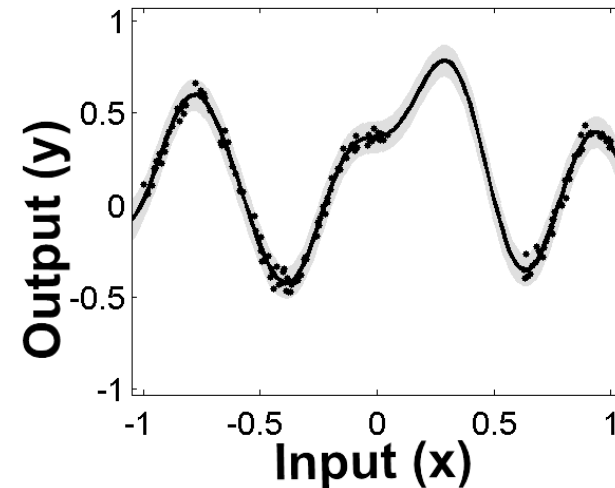
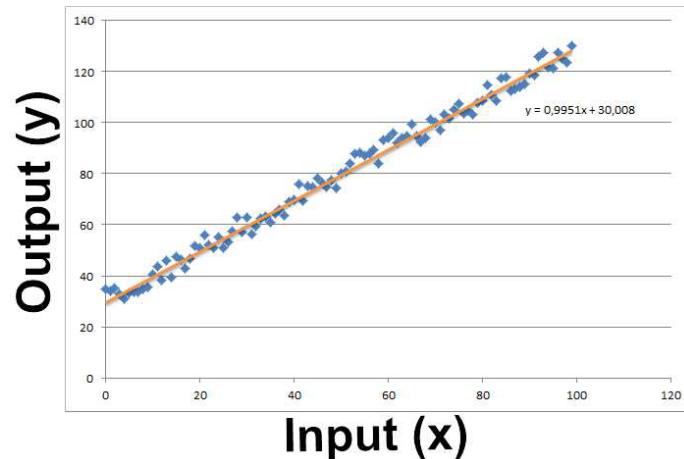


Real Valued Outputs: Regression

Housing price prediction.



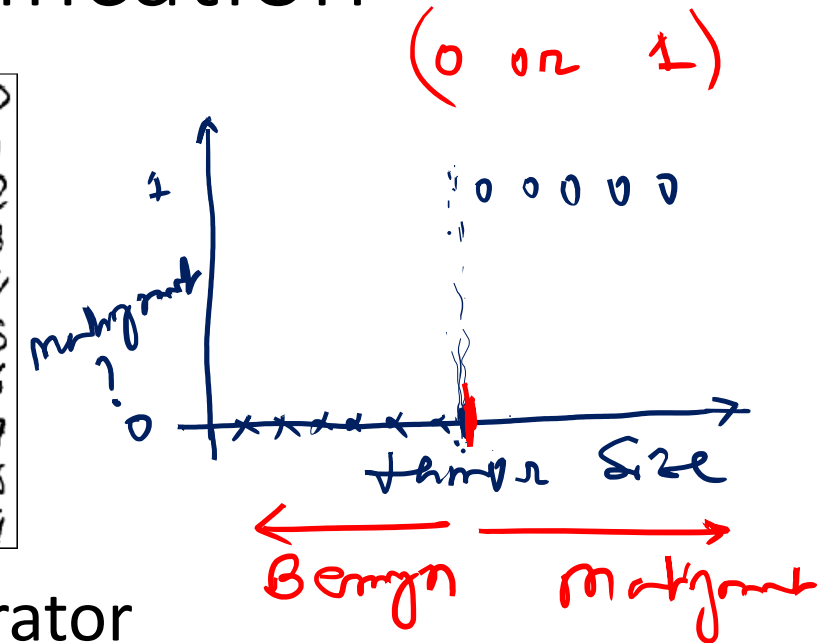
Regression: fitting a line/non-linear curve



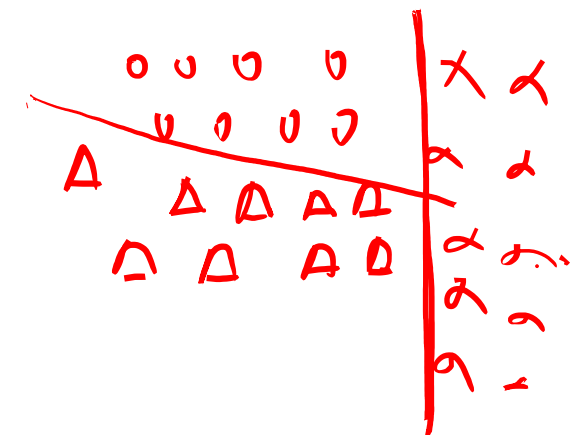
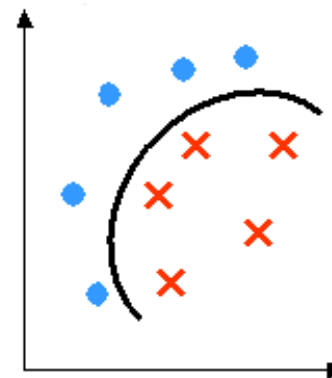
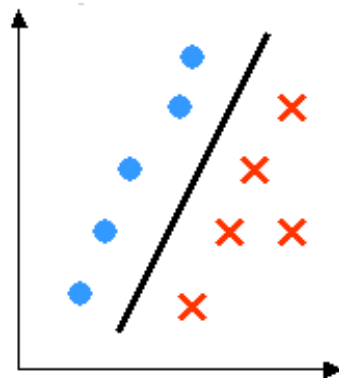
Discrete-valued outputs: Classification

(0-9)

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

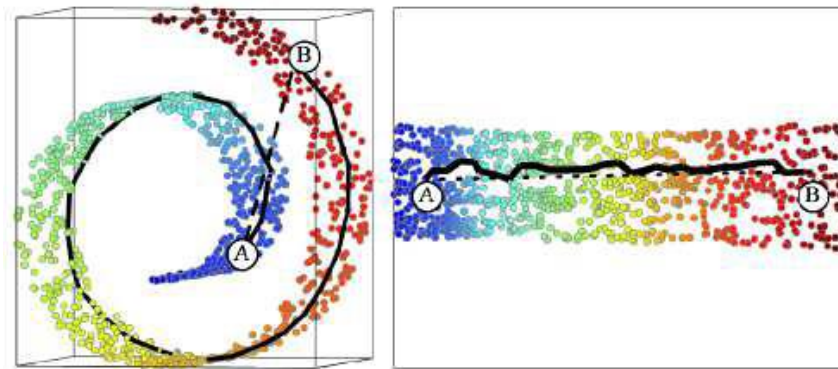
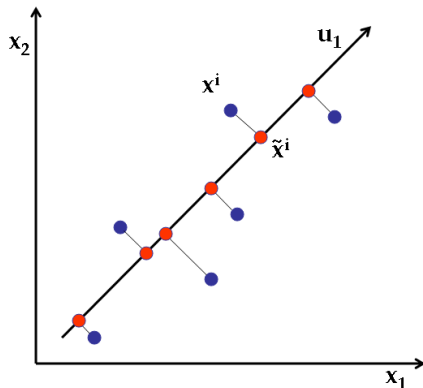
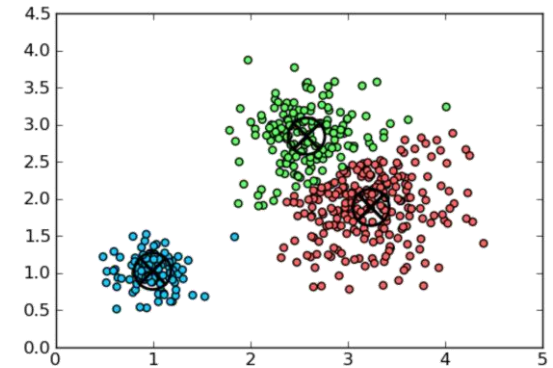


- Classification: finding a linear/nonlinear separator



Unsupervised Learning

- Given: Unlabelled training data $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$
- Objective: Learn the intrinsic latent structure that summarizes/explains data
- Homogeneous groups: clusters
- Low-dimensional latent structure: Dimensionality reduction

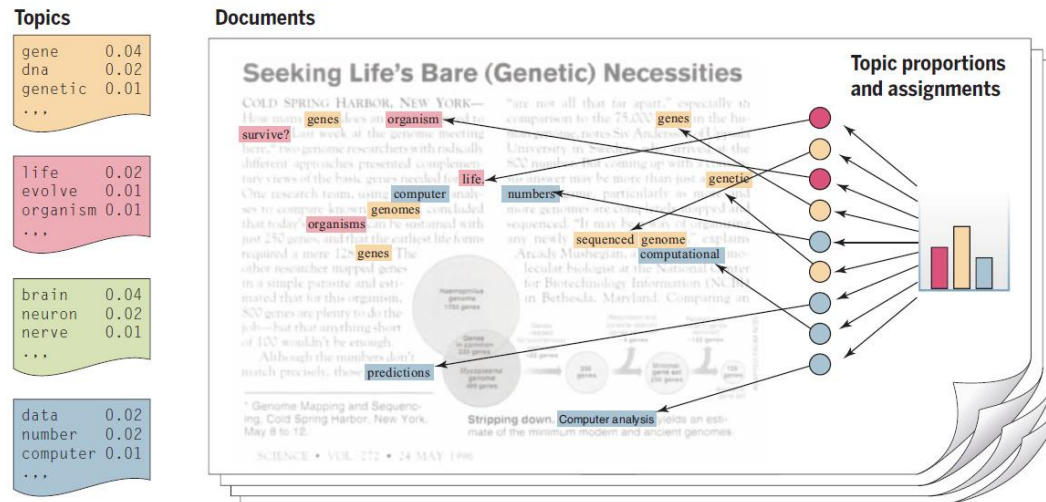


Unsupervised Learning: Some Examples

- Clustering large collections of images



- Topic discovery in large collections of text data



Topic modeling: D = Document

D1: The game was exciting and players performed well.

D2: The election results were surprising and the policy changes are impactful.

D3: The team scored well and won the match.

Latent Dirichlet Allocation (LDA)

Topic 1: Sports - { game, players, score, match, team, exciting
won }

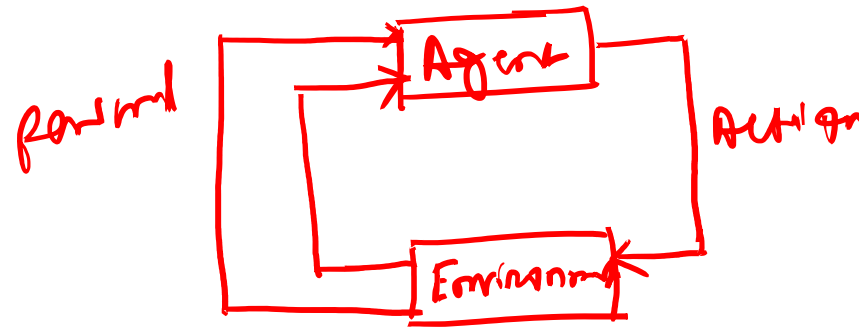
Topic 2: Politics - { election, results, policy, changes, impactful }

D1: 80% Sports, 20% Politics

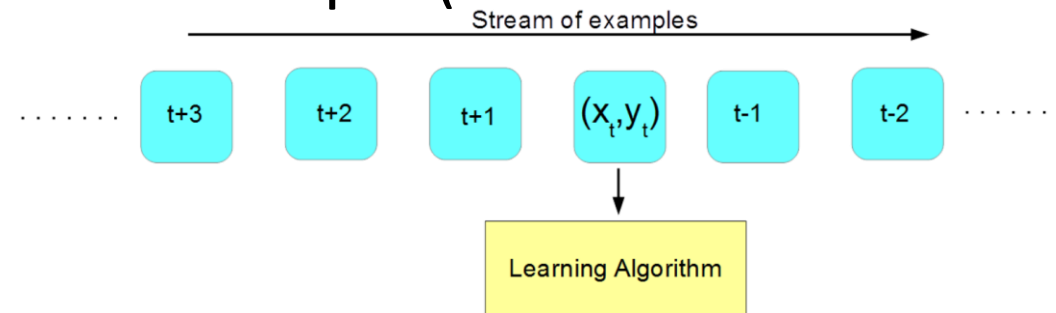
D2: 10% Sports, 90% Politics

D3: 95% " , 5% " .

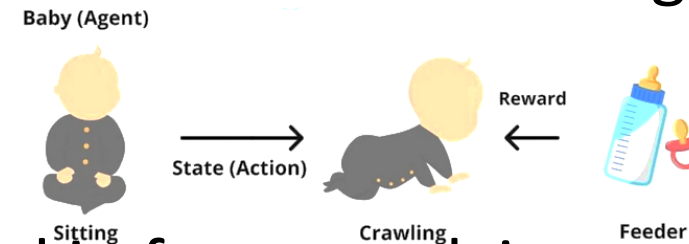
Others



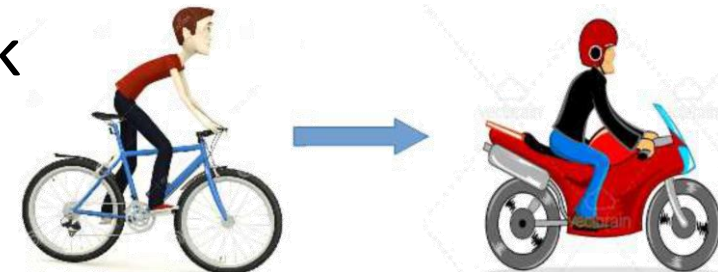
- Online learning: Learning with one example (or a small minibatch of examples) at a time



- Reinforcement Learning: Learning on trial and error through interactions with environment



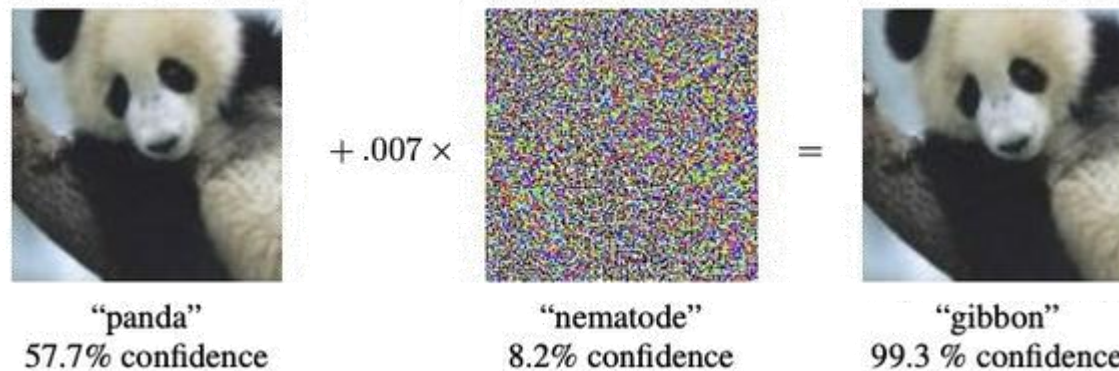
- Transfer Learning: Knowledge learned in from a task is re-used in order to boost performance on a related task



Data is the food for ML!

$$x_{adv} = x + \epsilon \cdot \text{Sign} \left(\nabla J(\theta; x) \right)$$

- What shall happen if the data is manipulated?



FGSM

- Adversarial machine learning is a machine learning method that aims to trick machine learning models by providing deceptive input.
- Attack: Modifications are performed on images that cause a classifier to produce incorrect predictions.
- Defense: Training model for defending such attacks.